258

B.Sc. Engg. CSE 1st Semester | 15 December 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION | **WINTER SEMESTER, 2021-2022**

DURATION: 3 HOURS | **FULL MARKS: 150**

CSE 4107: Structured Programming I

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

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1. a) There are 32 participating countries divided in four groups in the FIFA World Cup. Your task is to store the name of the teams group-wise along with the ranking of each team. 3×4 (CO1) (PO1)
   i. Write a structure named **"TeamInfo"** that can store the name and the ranking of a team.
   ii. Declare an array of that structure to store the group-wise information. (Figure 1 shows a pictorial representation of the array).
   iii. Write a program that reads the information and stores them into the array.

[team1, 13]	[team5, 2]	…	[team25, 1]	[team29, 20]
[team2, 14]	[team6, 6]	…	[team26, 3]	[team30, 22]
[team3, 12]	[team7, 8]	…	[team27, 4]	[team31, 17]
[team4, 15]	[team8, 5]	…	[team28, 9]	[team32, 23]

Figure1: Representation of array for Question 1.a)

   b) Based on the information stored in the question 1.a), build a scheduler using C program, that creates a group stage game schedule following round-robin procedure, where each team meets every other participant of their respective group. 13 (CO3) (PO2)

Consider the following criteria while creating the schedule:

- One game will be played every day without a break.
- First game will start on 01-Nov-2022.
- While printing the team's name for each of the games, the name of the team with better ranking (lower rank value) will be printed first.
- Follow the format provided in Figure 2 for printing the schedule.

```
Game 1: team1 vs team2 (01-Nov-2022)
Game 2: team3 vs team1 (02-Nov-2022)
Game 3: team1 vs team4 (03-Nov-2022)
```

Figure 2: Sample schedule of first 3 games

2. a) Due to the lack of space in the display board, the FIFA World Cup committee decided to display the shorter version of the player names. The names are shortened by the following guidelines: 10 (CO3) (PO2)

- Only the first name and last name are considered for the players.
- Full last name of the player with capitalized first letter is displayed first.
- Second part of the name contains capitalized first and last letter of first name.

For example:

- If a player's name is Mohammad salah, his name will be displayed as: Salah MD
- If a player's name is Lionel messi, his name will be displayed as: Messi LL

Construct a user-defined function that takes two strings as parameters representing the first and last name of a player. The function prints the shorter version of the player's name.

   b) Saudi Arabia made history in the FIFA World Cup 2022 by defeating Argentina. Being satisfied with the result, the prince of Saudi Arabia announced rewards for the players depending on the number of goals they scored in the tournament. The reward is calculated based on the following algorithm: 15 (CO3) (PO2)

259

```
1) input n (number of goals)
2) reward = n x 1000
3) repeat steps 4 to 6 if n is greater than 1
4)         if n is odd then n = 3n + 1
5)         else n = n/2
6)         reward = reward + ( n x 1000)
7) return reward
```

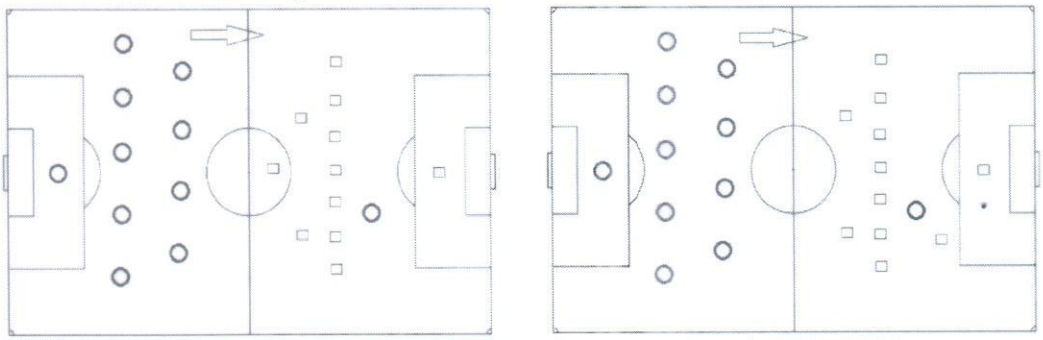
Figure 3: Algorithm for determining the reward for Question 2.(b)

For example:

- If anyone scored 4 goals, his reward will be:
  7000 = 4000 + 2000 + 1000
- If anyone scored 3 goals, his reward will be:
  49000 = 3000 + 10000 + 5000 + 16000 + 8000 + 4000 + 2000 + 1000

Your task is to construct a system using C program that reads the number of goals scored by a player and calculates his reward.

3. In a soccer game, an offside is a case when a player receives the ball from a teammate without having at least two opponent players between him and the goal. Figure 4 explains two scenarios representing an offside and a non-offside case. In these scenarios, the players of the two teams are represented using circles and squares. The arrow represents the direction of passing the ball. In both cases, the ball is sent from circle side to square side. 5+20 (CO3) (PO2)



Scenario 1: Offside Scenario 2: Not Offside

Figure 4: Offside scenarios for Question 3

- In the first scenario, there is only one square player between the last circle player and the goal post. That is why it is considered as offside.
- In the second scenario, there are two square players between the last circle player and the goal post. That is why it is not considered as offside.

You will be provided with the position of all 22 players of the field and the direction of the balls passed. The position of the players will be represented by a pair of numbers (x, y), a coordinate of a 2D plane. The bottom left corner of the field has a coordinate of (0,0) and the top-right corner of the field has a coordinate of (100,60). The input format is described below:

- The first line contains 11 pairs of numbers representing the position of the players of the circle side
- The second line contains 11 pairs of numbers representing the players from square side
- The third line contains a single character representing the direction of pass. A '>' represents the pass from circle side to square side and a '<' represents the pass from square side to circle side.

Answer the following questions:

a) Use a 3-dimensional array (2 x 11 x 2) to read the coordinates of the players into the array.
b) Construct an automated offside detector using C program, that given the input as described above, detects whether a certain pass is offside or not.
   A sample input and output is presented in Table 1.

260

Table 1: Sample input and output for Question 3

Input	Output
10,30 20,10 20,20 20,30 20,40 20,50 40,15 40,25 40,35 40,45 80,20 55,30 60,10 60,40 65,15 65,20 65,25 65,30 65,35 65,40 65,45 95,30 >	Offside
10,30 20,10 20,20 20,30 20,40 20,50 40,15 40,25 40,35 40,45 80,20 85,15 60,10 60,40 65,15 65,20 65,25 65,30 65,35 65,40 65,45 95,30 >	Not Offside

4. a) Consider the code snippet provided in Figure 5: 3×4 (CO3) (PO2)

```
#include<stdio.h>
void f1(char *str){
    if(*str){
        printf("%c",*str);
        //Recursive Call
        printf("%c",*str);
    }
}
int main(){
    f1("FIFA-2022");
}
```

Figure 5: Code snippet for question 4.a)

Find the output of the program if you replace the following statements in the place of "`// Recursive Call`":

i. `f1(str+1);`
ii. `f1(str++);`
iii. `f1(++str);`

b) To create a specific pattern using the spectators in 2022 World Cup, FIFA decided to allocate seats to specific fan base maintaining the following criteria: 13 (CO3) (PO2)

- If the topmost row contains n spectators, then the next row will contain $(n-1)$ spectators
- Once the spectator number reaches to 1, then it starts increasing by 1 in the following rows until it reaches n spectators

A sample seat allocation where top row contains 5 spectators is presented in Figure 6.

```
#####
####
###
##
#
##
###
####
#####
```

Figure 6: Sample seat distribution for question 4.b)

Construct a recursive function that takes a number n, indicating the number of seats in the top row and prints the seat distribution pattern.

5. a) Assume that you want to build an information system using files to store some basic information about each team in the FIFA World Cup. The system needs to be updated after every match. The following information will be stored in the system: 15 (CO3) (PO2)

- Team name
- Team rank
- Total number of red cards
- Total number of yellow cards

261

Construct the information system using C program, having the following features:

- The system stores the information for all 32 countries.
- The system loads the previously written information from a file named "info.txt". You can assume a preloaded file is available with the team names and ranks. The red and yellow card counts are set to 0 initially.
- The system asks the user about the number of red and yellow cards of each team from the latest match and write the updated information back to the same file.

b) Toffee is a streaming platform which is the official broadcaster of FIFA World Cup 2022 in Bangladesh. To create an account in toffee, users need to set their username and password. The password must meet some criteria to be a valid password. The criteria are provided below: 10 (CO3) (PO2)

- The length of the password must be at least 10
- The password must contain both uppercase and lowercase letter
- The password must contain at least one number
- The password cannot have the word "password" as a part of it

Your task is to construct a password validator function using C program that takes a string as a parameter representing a password provided by the user. The function returns 1, if the password is valid, otherwise it returns 0.

6. a) Write a C program that prints its own source code n number of times, where n is a command line argument passed to the main function. 13 (CO2) (PO2)
[Hint: In C, the macro **__FILE__** holds the path to the current file.]

b) The code snippet provided in the Figure 7 is an attempt to implement a library function. There are some mistakes in the implementation. 2+5 (CO2) (PO1)

```
char* f1(char* a, char* b){
    char *begin = *a;
    while(*a)
        a++;
    while(*b)
        *a++ = *b++;
    *a = '\0';
    return begin;
}
```

Figure 7: Code snippet for question 6.b)

i. Identify the library function was attempted to be implemented.
ii. Convert the code by correcting the mistakes in the given code snippet. (Any unnecessary changes will cause penalty)

c) Is it possible to design a user-defined function which takes a number as a parameter and returns both of its square and square root? If your answer is no, explain why. If your answer is yes, provide a solution. 5 (CO2) (PO1)